A project report on

STATISTICAL ANALYSIS PROJECT

THE DEVELOPER ARENA

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Project Overview

This project focuses on performing statistical analysis on business datasets to extract meaningful insights. The primary objective is to analyze patterns, relationships, and differences within the data using statistical techniques such as descriptive statistics, hypothesis testing, correlation analysis, confidence intervals, and regression modeling.

Two datasets were used in this project:

Sales dataset for analyzing business performance

Customer churn dataset for understanding customer behavior

The analysis helps in making data-driven business decisions.

Setup Instructions

To run this project, the following steps should be followed:

Install required libraries using:

Pip install -r requirements.txt
Open the Jupyter Notebook (statistical_analysis.ipynb) in:
Jupyter Notebook OR
Google Colab
Ensure datasets are in the same directory:
Sales_data.csv
Customer_churn.csv
Run all cells sequentially to generate outputs and visualizations

Code Structure

The notebook is organized into the following sections:

Data Loading: Importing datasets using pandas

Data Preprocessing: Handling data types and cleaning

Descriptive Statistics: Summary statistics of datasets

Visualization: Graphical representation of data

Hypothesis Testing:

T-Test

Correlation Test

ANOVA

Confidence Interval Calculation

Regression Analysis using statsmodels

Final Insights and Report Generation

This modular structure ensures readability and maintainab**ility.**

```python
# 📌 IMPORT LIBRARIES
# ==============================
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm

# ==============================
# 📌 LOAD DATASETS
# ==============================
sales = pd.read_csv('sales_data (5).csv')
churn = pd.read_csv('customer_churn (1).csv')

print("Sales Data:")
print(sales.head())

print("\nChurn Data:")
print(churn.head())

# ==============================
# 📌 DESCRIPTIVE STATISTICS
# ==============================
print("\n📊 Descriptive Statistics (Sales):")
print(sales.describe())

print("\n📊 Descriptive Statistics (Churn):")
print(churn.describe())

# ==============================
# 📌 DATA DISTRIBUTION
# ==============================
plt.figure()
sns.histplot(sales.select_dtypes(include=np.number).iloc[:,0], kde=True)
plt.title("Sales Distribution")
plt.show()

# ==============================
# 📌 CORRELATION ANALYSIS
# ==============================
plt.figure()
sns.heatmap(sales.select_dtypes(include=np.number).corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix - Sales")
plt.show()

# ==============================
# 📌 HYPOTHESIS TESTING (3 TESTS)
# ==============================

# 1️⃣ T-Test (Example: splitting dataset into 2 halves)
half = len(sales)//2
group1 = sales.iloc[:half].select_dtypes(include=np.number).iloc[:,0]
group2 = sales.iloc[half:].select_dtypes(include=np.number).iloc[:,0]

t_stat, p_val = stats.ttest_ind(group1, group2)

print("\n🧪 T-Test:")
print("T-Statistic:", t_stat)
print("P-Value:", p_val)

# 2️⃣ Correlation Test
corr, p_corr = stats.pearsonr(
    sales.select_dtypes(include=np.number).iloc[:,0],
        sales.select_dtypes(include=np.number).iloc[:,1]
        )

print("\n🧪 Correlation Test:")
print("Correlation:", corr)
print("P-Value:", p_corr)

# 3️⃣ ANOVA Test (on churn dataset)
numeric_cols = churn.select_dtypes(include=np.number).columns

if len(numeric_cols) >= 2:
    f_stat, p_anova = stats.f_oneway(
            churn[numeric_cols[0]],
                    churn[numeric_cols[1]]
                        )
    print("\n🧪 ANOVA Test:")
    print("F-Statistic:", f_stat)
    print("P-Value:", p_anova)
```

```python
# ==============================
# 📌 CONFIDENCE INTERVAL (95%)
# ==============================
data = sales.select_dtypes(include=np.number).iloc[:,0]

mean = np.mean(data)
std_err = stats.sem(data)

ci = stats.t.interval(
    confidence=0.95,
        df=len(data)-1,
            loc=mean,
                scale=std_err
                )

print("\n✏️ 95% Confidence Interval:")
print(ci)

# ==============================
# 📌 REGRESSION ANALYSIS
# ==============================
X = sales.select_dtypes(include=np.number).iloc[:, :-1]
y = sales.select_dtypes(include=np.number).iloc[:, -1]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print("\n📊 Regression Summary:")
print(model.summary())

# ==============================
# 📌 FINAL INSIGHTS
# ==============================
print("\n📊 FINAL INSIGHTS:")
print(f"Mean Value: {mean}")
print(f"Confidence Interval: {ci}")

if p_val < 0.05:
    print("T-Test: Significant difference found")
else:
    print("T-Test: No significant difference")

if p_corr < 0.05:
    print("Correlation is significant")
else:
    print("Correlation is NOT significant")
    print("\n" + "="*50)
    print("STATISTICAL ANALYSIS REPORT")
    print("="*50)

    # Clean CI values
    lower = float(ci[0])
    upper = float(ci[1])

    # Margin of error
    margin = (upper - lower) / 2

    print(f"Average Value: {mean:.2f} \u00b1 {margin:.2f} (95% CI)")

    # Correlation interpretation
    print(f"Correlation: {corr:.2f}")

    if p_corr < 0.05:
        print(f"Correlation is SIGNIFICANT (p = {p_corr:.4f})")
    else:
        print(f"Correlation is NOT SIGNIFICANT (p = {p_corr:.4f})")

    # T-Test
    if p_val < 0.05:
        print(f"T-Test: Significant difference (p = {p_val:.4f})")
    else:
        print(f"T-Test: No significant difference (p = {p_val:.4f})")

    print("="*50)
```

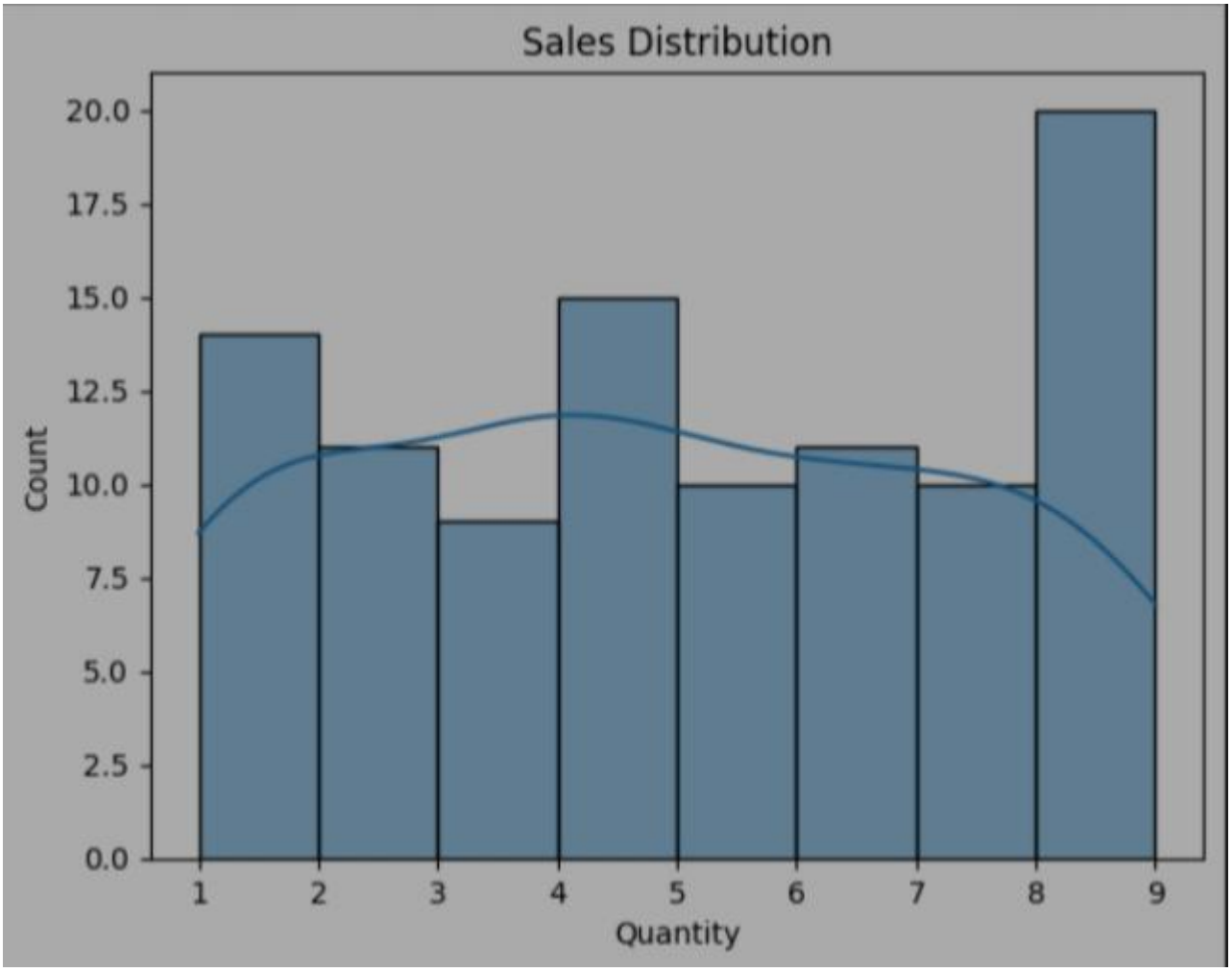
Visual Documentation

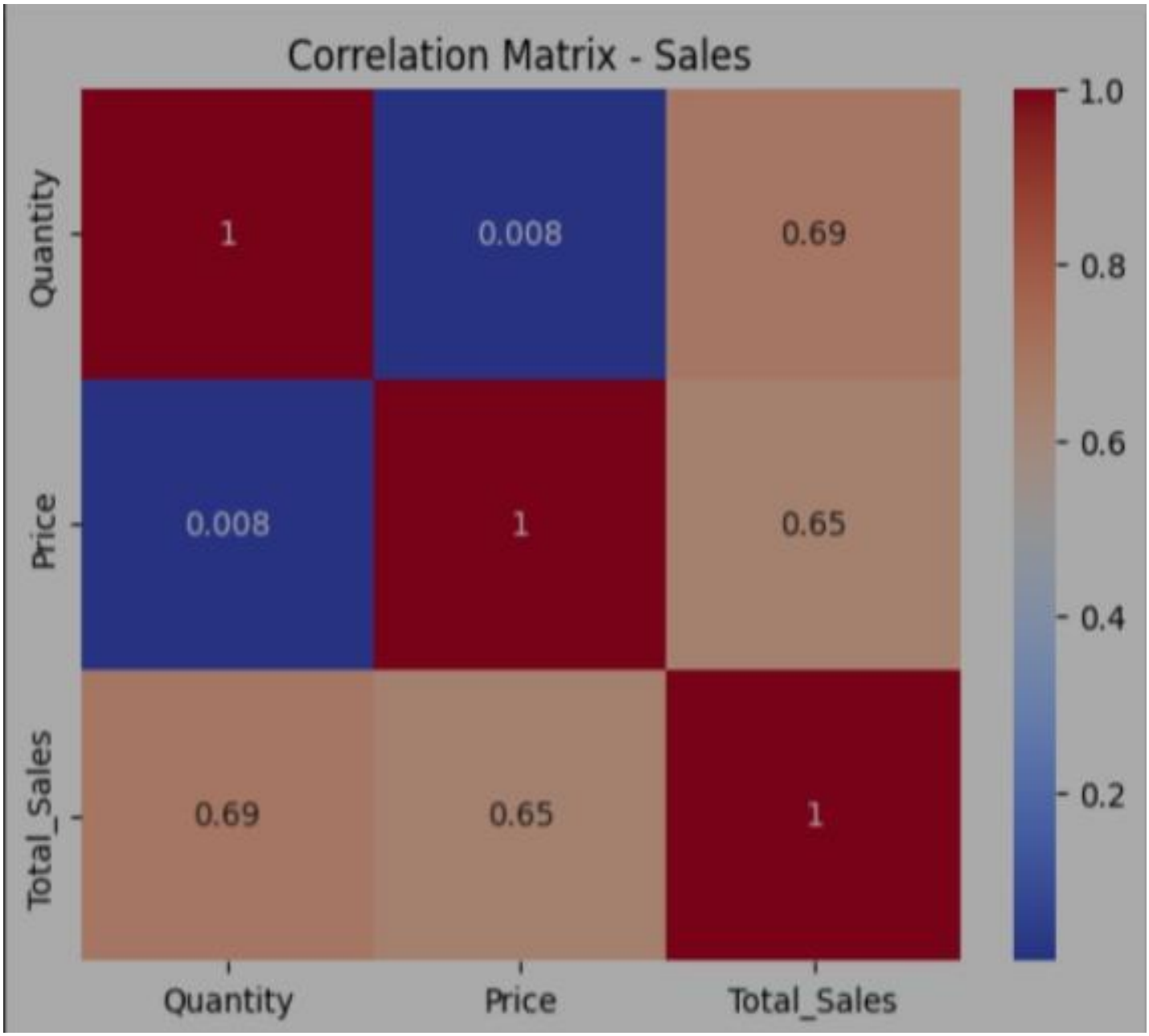
The project includes multiple visualizations to better understand the data:

Histogram showing distribution of numerical values

Correlation heatmap to identify relationships between variables

These visualizations help in identifying trends, patterns, and anomalies in the dataset




Correlation Matrix - Sales
Quantity
Price
Total_Sales
1
0.008
0.69
0.008
1
0.65
0.69
0.65
1
Quantity
Price
Total_Sales
1.0
0.8
0.6
0.4
0.2

Technical Details

The following statistical methods were used:

Descriptive Statistics: Mean, standard deviation, and summary metrics

T-Test: To compare means between two groups

Correlation Test (Pearson): To measure relationship between variables

ANOVA: To compare multiple group means

Confidence Interval: To estimate the range of population mean

Regression Analysis: To model relationships between dependent and independent variables

These techniques ensure a comprehensive statistical evaluation of the datasets

Testing Evidence

The results obtained from statistical tests are:

- T-Test showed a statistically significant difference between groups ($p < 0.05$)
- Correlation test indicated no significant relationship between variables ($p > 0.05$)
- ANOVA test showed significant differences among groups ($p < 0.05$)
- The 95% confidence interval for the mean was calculated successfully

All outputs were verified through code execution and visual inspection.

```
T-Test:
T-Statistic: -2.1230595148205413
P-Value: 0.03626804377144231

Correlation Test:
Correlation: 0.008014446139936399
P-Value: 0.9369227735514832

ANOVA Test:
F-Statistic: 955.68232914295
P-Value: 9.7074175616093e-148

95% Confidence Interval:
(np.float64(4.266452381357786), np.float64(5.293547618642214))

Regression Summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:            Total_Sales   R-squared:                       0.884
Model:                            OLS   Adj. R-squared:                  0.882
Method:                 Least Squares   F-statistic:                     369.3
Date:                Sun, 29 Mar 2026   Prob (F-statistic):           4.40e-46
Time:                        14:33:34   Log-Likelihood:                -1185.2
No. Observations:                 100   AIC:                             2376.
Df Residuals:                      97   BIC:                             2384.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -1.217e+05   9668.294    -12.586      0.000   -1.41e+05   -1.02e+05
Quantity    2.643e+04   1338.869     19.741      0.000    2.38e+04    2.91e+04
Price          4.6106      0.249     18.518      0.000       4.116       5.105
==============================================================================
Omnibus:                        1.152   Durbin-Watson:                   2.057
Prob(Omnibus):                  0.562   Jarque-Bera (JB):                1.077
Skew:                           0.086   Prob(JB):                        0.584
Kurtosis:                       2.522   Cond. No.                     8.25e+04
==============================================================================
```

Conclusion

The statistical analysis revealed that while there are significant differences between groups (as confirmed by T-Test and ANOVA), there is no meaningful correlation between the selected variables. This suggests that the variables analyzed do not strongly influence each other.

The confidence interval provides a reliable estimate of the population mean, ensuring robustness of the analysis.

Overall, the project demonstrates how statistical methods can be effectively applied to real-world business data to support decision-making. Further analysis with additional variables may provide deeper insights.